AMATH/PMATH 331: Applied Real Analysis Fall 2026

Assignment 2; Solutions

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[1] Let $(\mathbf{x}_k)_{k\in\mathbb{N}}$ be a sequence in $\mathbb{R}^n$. Write $\mathbf{x}_k = (x_{k1}, \ldots, x_{kn})$, so that $(x_{kj})_{k\in\mathbb{N}}$ is a sequence in $\mathbb{R}$ for each $j = 1, \ldots, n$.

[a] Let $\mathbf{a} = (a_1, \ldots, a_n) \in \mathbb{R}^n$. Prove that $\lim_{k\to\infty} \mathbf{x}_k = \mathbf{a}$ if and only if $\lim_{k\to\infty} x_{kj} = a_j$ for each $j = 1, \ldots, n$. (*Don't forget to prove both directions!*)

[b] Prove that $(\mathbf{x}_k)_{k\in\mathbb{N}}$ is a Cauchy sequence in $\mathbb{R}^n$ if and only if $(x_{kj})_{k\in\mathbb{N}}$ is a Cauchy sequence in $\mathbb{R}$ for each $j = 1, \ldots, n$. (*Don't forget to prove both directions!*)

Solution.

[a] Suppose $\lim_{k\to\infty} \mathbf{x}_k = \mathbf{a}$. Let $\varepsilon > 0$. There exists $N \in \mathbb{N}$ such that $k \geq N$ implies $\|\mathbf{x}_k - \mathbf{a}\|^2 < \varepsilon^2$. But then

$$\sum_{j=1}^{n} |x_{kj} - a_j|^2 < \varepsilon^2,$$

so $|x_{kj} - a_j| < \varepsilon$ for all $k \geq N$. This says $\lim_{k\to\infty} x_{kj} = a_j$.

Conversely, suppose $\lim_{k\to\infty} x_{kj} = a_j$ for each $j = 1, \ldots, n$. Let $\varepsilon > 0$. For each $j = 1, \ldots, n$, there exists $N_j \in \mathbb{N}$ such that $k \geq N_j$ implies that $|x_{kj} - a_j|^2 < \frac{1}{n}\varepsilon^2$. Let $N = \max_{j=1}^n N_j$. Then if $k \geq N$, we have

$$\|\mathbf{x}_k - \mathbf{a}\|^2 = \sum_{j=1}^{n} |x_{kj} - a_j|^2 < \sum_{j=1}^{n} \frac{1}{n}\varepsilon^2 = \varepsilon^2,$$

so $\|\mathbf{x}_k - \mathbf{a}\| < \varepsilon$ for all $k \geq N$. This says $\lim_{k\to\infty} \mathbf{x}_k = \mathbf{a}$.

[b] Suppose that $(\mathbf{x}_k)_{k\in\mathbb{N}}$ is Cauchy. Let $\varepsilon > 0$. There exists $N \in \mathbb{N}$ such that $k, l \geq N$ implies $\|\mathbf{x}_k - \mathbf{x}_l\|^2 < \varepsilon^2$. But then

$$\sum_{j=1}^{n} |x_{kj} - x_{lj}|^2 < \varepsilon^2,$$

so $|x_{kj} - x_{lj}| < \varepsilon$ for all $k, l \geq N$. This says $(x_{kj})_{k\in\mathbb{N}}$ is Cauchy for each $j = 1, \ldots, n$.

Conversely, suppose $(x_{kj})_{k\in\mathbb{N}}$ is Cauchy for each $j = 1, \ldots, n$. For each $j = 1, \ldots, n$, there exists $N_j \in \mathbb{N}$ such that $k, l \geq N_j$ implies that $|x_{kj} - x_{lj}|^2 < \frac{1}{n}\varepsilon^2$. Let $N = \max_{j=1}^n N_j$. Then if $k, l \geq N$, we have

$$\|\mathbf{x}_k - \mathbf{x}_l\|^2 = \sum_{j=1}^{n} |x_{kj} - x_{lj}|^2 < \sum_{j=1}^{n} \frac{1}{n}\varepsilon^2 = \varepsilon^2,$$

so $\|\mathbf{x}_k - \mathbf{x}_l\| < \varepsilon$ for all $k, l \geq N$. This says $(x_{kj})_{k\in\mathbb{N}}$ is Cauchy.

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[2] The two parts of this problem *are* related.

[a] Let $E \subseteq \mathbb{R}$ be nonempty. If E is bounded below, so that $\inf E$ exists, prove that there exists a sequence (a_k) with $a_k \in E$ for all k, and $\lim_{k\to\infty} a_k = \inf E$.

[b] Let $A \subseteq \mathbb{R}^n$ be nonempty, and $\mathbf{x} \in \mathbb{R}^n$. We define the *distance* from A to $\mathbf{x}$ to be

$$\operatorname{dist}_A(\mathbf{x}) = \inf\{\|\mathbf{x} - \mathbf{y}\| : \mathbf{y} \in A\}.$$

(Note that $\operatorname{dist}_A(\mathbf{x})$ is well-defined and $\operatorname{dist}_A(\mathbf{x}) \geq 0$, because it is the infimum of a nonempty set of nonnegative real numbers.) Prove that there exists a sequence $(\mathbf{y}_k)$ in $\mathbb{R}^n$ with $\mathbf{y}_k \in A$ for all k and $\lim_{k\to\infty} \|\mathbf{x} - \mathbf{y}_k\| = \operatorname{dist}_A(\mathbf{x})$.

Solution.

[a] If $k \in \mathbb{N}$, then $\inf E + \frac{1}{k}$ is not a lower bound for E, so there exists $a_k \in E$ with

$$\inf E \leq a_k < \inf E + \frac{1}{k}.$$

Then by the squeeze theorem we have $\lim_{k\to\infty} a_k = \inf E$.

[b] Let $E = \{\|\mathbf{x} - \mathbf{y}\| : \mathbf{y} \in A\}$, which is nonempty and bounded below. By part [a], there exists a sequence (a_k) in E such that (a_k) converges to $\inf E = \operatorname{dist}_A(\mathbf{x})$. But then $a_k = \|\mathbf{x}-\mathbf{y}_k\|$ for some $\mathbf{y}_k \in A$. So we have a sequence $(\mathbf{y}_k)$ in $\mathbb{R}^n$ with $\mathbf{y}_k \in A$ for all k and $\lim_{k\to\infty} \|\mathbf{x}-\mathbf{y}_k\| = \operatorname{dist}_A(\mathbf{x})$.

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[3] Let $A \subseteq \mathbb{R}^n$. We define the *closure* of A, denoted by $\overline{A}$, to be the subset

$$\overline{A} = \{\mathbf{x} \in \mathbb{R}^n : B_\varepsilon(\mathbf{x}) \cap A \neq \varnothing \text{ for all } \varepsilon > 0\}.$$

That is, a point $\mathbf{x} \in \mathbb{R}^n$ lies in the closure of A if and only if every open ball of any positive radius intersects A.

[a] Show that $A \subseteq \overline{A}$.

[b] Show that $\overline{A} = \{\mathbf{x} \in \mathbb{R}^n : U \cap A \neq \varnothing \text{ for any open set } U \text{ containing } \mathbf{x}\}$.

[c] Show that $\overline{A}$ is closed.

Hint: Let $\mathbf{x} \in (\overline{A})^c$. First show there exists ε_0 such that $B_{\varepsilon_0}(\mathbf{x}) \cap A = \varnothing$, and then show that $B_{\varepsilon_0}(\mathbf{x}) \cap \overline{A} = \varnothing$. You will need to use part [b].

[d] If A is closed, show that $\overline{A} = A$.

(Combining parts [c] and [d] shows that A is closed if and only if $\overline{A} = A$.)

Solution.

[a] Let $\mathbf{x} \in A$. For any $\varepsilon > 0$, since $\mathbf{x} \in B_\varepsilon(\mathbf{x})$, we have $B_\varepsilon(\mathbf{x}) \cap A \neq \varnothing$. Thus $\mathbf{x} \in \overline{A}$, so $A \subseteq \overline{A}$.

[b] To simplify notation, write

$$\overline{A} = \{\mathbf{x} \in \mathbb{R}^n : B_\varepsilon(\mathbf{x}) \cap A \neq \varnothing \text{ for all } \varepsilon > 0\},$$
$$P := \{\mathbf{x} \in \mathbb{R}^n : U \cap A \neq \varnothing \text{ for any open set } U \text{ containing } \mathbf{x}\}.$$

We want to show that $\overline{A} = P$. If $\mathbf{x} \in P$, then $U \cap A \neq \varnothing$ for any open set U containing $\mathbf{x}$. This includes all open balls $B_\varepsilon(\mathbf{x})$ centred at $\mathbf{x}$, so $\mathbf{x} \in \overline{A}$. Thus $P \subseteq \overline{A}$.

Conversely, suppose $\mathbf{x} \in \overline{A}$. Let U be an open set containing $\mathbf{x}$. Then there exists $\varepsilon > 0$ such that $B_\varepsilon(\mathbf{x}) \subseteq U$, so $U \cap A$ contains $B_\varepsilon(\mathbf{x}) \cap A$ which is nonempty. Thus $U \cap A \neq \varnothing$ for any such U, so $\mathbf{x} \in P$ and hence $\overline{A} \subseteq P$.

[c] Let $\mathbf{x} \in (\overline{A})^c$. By definition of $\overline{A}$, this means there exists some $\varepsilon_0 > 0$ such that $B_{\varepsilon_0}(\mathbf{x}) \cap A = \varnothing$. We claim that in fact $B_{\varepsilon_0}(\mathbf{x}) \cap \overline{A} = \varnothing$. To se this, suppose that $B_{\varepsilon_0}(\mathbf{x}) \cap \overline{A} \neq \varnothing$, so there exists $\mathbf{y} \in B_{\varepsilon_0} \cap \overline{A}$. But this means that $\mathbf{y} \in \overline{A}$, so by part [b], $U \cap A \neq \varnothing$ for any open set containing $\mathbf{y}$. Since $B_{\varepsilon_0}(\mathbf{x})$ is an open set containing $\mathbf{y}$, this contradicts the fact that $B_{\varepsilon_0}(\mathbf{x}) \cap A = \varnothing$. Thus we must have $B_{\varepsilon_0}(\mathbf{x}) \cap \overline{A} = \varnothing$, so $B_{\varepsilon_0}(\mathbf{x}) \subseteq (\overline{A})^c$. Since $\mathbf{x} \in (\overline{A})^c$ was arbitrary, we deduce that $(\overline{A})^c$ is open, and thus $\overline{A}$ is closed.

[d] Suppose A is closed. We want to show that $\overline{A} = A$. Since we always have $A \subseteq \overline{A}$, it remains to show that $\overline{A} \subseteq A$. Suppose not. Then there exists $\mathbf{x} \in \overline{A}$ with $\mathbf{x} \notin A$, so $\mathbf{x} \in A^c$. Since A^c is open, there exists $\varepsilon_0 > 0$ such that $B_{\varepsilon_0}(\mathbf{x}) \subseteq A^c$, so $B_{\varepsilon_0} \cap A = \varnothing$. But this contradicts that $\mathbf{x} \in \overline{A}$.

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[4] Let $E \subseteq \mathbb{R}^n$ be nonempty.

[a] If $(\mathbf{x}_k)_{k\in\mathbb{N}}$ is a sequence such that $\mathbf{x}_k \in E$ for all $k \in \mathbb{N}$, and $\lim_{k\to\infty} \mathbf{x}_k = \mathbf{p}$, prove that $\mathbf{p} \in \overline{E}$.

[b] Let $\mathbf{p} \in \overline{E}$. Prove that there exists a sequence $(\mathbf{x}_k)_{k\in\mathbb{N}}$ such that $\mathbf{x}_k \in E$ for all $k \in \mathbb{N}$, and $\lim_{k\to\infty} \mathbf{x}_k = \mathbf{p}$. *Hint:* Consider $B_{\frac{1}{k}}(\mathbf{p})$.

Taken together, these two statements say that $\overline{E}$ is exactly the set of limits of convergent sequences whose elements all lie in E.

Solution.

[a] Let $\delta > 0$. Since $\lim_{k\to\infty} \mathbf{x}_k = \mathbf{p}$, there exists $N \in \mathbb{N}$ such that $k \geq N$ implies $\mathbf{x}_k \in B_\delta(\mathbf{a})$. But then $B_\delta(\mathbf{p}) \cap E \neq \varnothing$. Since $\delta > 0$ is arbitrary, we conclude that $\mathbf{a} \in \overline{E}$.

[b] Let $\mathbf{p} \in \overline{E}$. Let $k \in \mathbb{N}$, so $\frac{1}{k} > 0$. Then $B_{\frac{1}{k}}(\mathbf{p}) \cap E \neq \varnothing$. So we can choose $\mathbf{x}_k \in B_{\frac{1}{k}}(\mathbf{p}) \cap E$. In particular, $\mathbf{x}_k \in E$ for all $k \in \mathbb{N}$. Moreover, given $\varepsilon > 0$, if we take $N \in \mathbb{N}$ such that $\frac{1}{N} < \varepsilon$, then $k \geq N$ implies $\frac{1}{k} \leq \frac{1}{N} < \varepsilon$, so $\mathbf{x}_k \in B_{\frac{1}{k}}(\mathbf{p}) \subseteq B_\varepsilon(\mathbf{p})$. This says $\lim_{k\to\infty} \mathbf{x}_k = \mathbf{p}$.

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[5] Let $(\mathbf{x}_m)_{m\in\mathbb{N}}$ be a sequence in $\mathbb{R}^n$. Define a new sequence $(\mathbf{s}_k)_{k\in\mathbb{N}}$ by

$$\mathbf{s}_k = \sum_{m=1}^{k} \mathbf{x}_m.$$

If the sequence $(\mathbf{s}_k)$ converges, then we say that *the series* $\sum_{m=1}^{\infty} \mathbf{x}_m$ *converges*. If $\lim_{k\to\infty} \mathbf{s}_k = \mathbf{s}$, then we write $\sum_{m=1}^{\infty} \mathbf{x}_m = \mathbf{s}$.

[a] Prove that $\sum_{m=1}^{\infty} \mathbf{x}_m$ converges iff: for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$\left\| \sum_{m=k+1}^{\ell} \mathbf{x}_m \right\| < \varepsilon$$

for all $\ell \geq k \geq N$.

Hint: A sequence in $\mathbb{R}^n$ converges if and only if it is Cauchy.

[b] Suppose that $\sum_{m=1}^{\infty} \mathbf{x}_m$ converges. For any $N \in \mathbb{N}$, prove that $\sum_{m=N}^{\infty} \mathbf{x}_m$ converges, and that

$$\lim_{N\to\infty} \left(\sum_{m=N}^{\infty} \mathbf{x}_m \right) = 0.$$

Hint: Consider the sequence $(\tilde{\mathbf{s}}_\ell)_{\ell\geq N}$ given by $\tilde{\mathbf{s}}_\ell = \sum_{m=N}^{\ell} \mathbf{x}_m$.

Solution.

[a] A sequence in $\mathbb{R}^n$ converges if and only if it is Cauchy. Recall that saying $\sum_{m=1}^{\infty} \mathbf{x}_m$ converges means that the sequence $(\mathbf{s}_k)$ converges, where $\mathbf{s}_k = \sum_{m=1}^{k} \mathbf{x}_m$. This is equivalent to $(\mathbf{s}_k)$ being Cauchy, which says that for all $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $\ell \geq k \geq N$ implies

$$|\mathbf{s}_\ell - \mathbf{s}_k| = \left\| \sum_{m=k+1}^{\ell} \mathbf{x}_m \right\| < \varepsilon.$$

[b] Suppose that $\sum_{m=1}^{\infty} \mathbf{x}_m$ converges, which means that $(\mathbf{s}_k)$ converges, where $\mathbf{s}_k = \sum_{m=1}^{k} \mathbf{x}_m$. Let $\mathbf{s} = \lim_{k\to\infty} \mathbf{s}_k$. Fix $N \in \mathbb{N}$, and define a new sequence $(\tilde{\mathbf{s}}_\ell)_{\ell\geq N}$ by $\tilde{\mathbf{s}}_\ell = \sum_{m=N}^{\ell} \mathbf{x}_m$. Since $\tilde{\mathbf{s}}_\ell = \mathbf{s}_l - \sum_{m=1}^{N-1} \mathbf{x}_m$, we have

$$|\tilde{\mathbf{s}}_k - \tilde{\mathbf{s}}_\ell| = |\mathbf{s}_k - \mathbf{s}_\ell|,$$

so $(\tilde{\mathbf{s}}_\ell)_{\ell\geq N}$ is Cauchy if and only if $(\mathbf{s}_k)_{k\in\mathbb{N}}$ is Cauchy, which it is because we assume $(\mathbf{s}_k)$ converges. Thus $(\tilde{\mathbf{s}}_\ell)_{\ell\geq N}$ converges, which can be equivalently expressed as saying that $\sum_{m=N}^{\infty} \mathbf{x}_m$ converges.

By the definition of $(\mathbf{s}_k)$, if $k \geq N$ we have

$$\mathbf{s}_k = \sum_{m=1}^{k} \mathbf{x}_m = \sum_{m=1}^{N-1} \mathbf{x}_m + \sum_{m=N}^{k} \mathbf{x}_m = \mathbf{s}_{N-1} + \sum_{m=N}^{k} \mathbf{x}_m.$$

Taking the limit as $k \to \infty$ above gives $\mathbf{s} = \mathbf{s}_{N-1} + \sum_{m=N}^{\infty} \mathbf{x}_m$. Since $\lim_{N\to\infty} \mathbf{s}_{N-1} = \mathbf{s}$, we then get $\lim_{N\to\infty}(\sum_{m=N}^{\infty} \mathbf{x}_m) = 0$.

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[6] Let $(\mathbf{x}_k)_{k\in\mathbb{N}}$ be a sequence in $\mathbb{R}^n$. Given $\mathbf{x} \in \mathbb{R}^n$, prove that $(\mathbf{x}_k)$ has a *subsequence* converging to $\mathbf{x}$ if and only if, for all $\varepsilon > 0$ the open ball $B_\varepsilon(\mathbf{x})$ contains $\mathbf{x}_k$ for infinitely many k.

Solution. Suppose first that $(\mathbf{x}_k)$ has a subsequence $(\mathbf{x}_{f(k)})$ converging to $\mathbf{x}$. For any $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $k \geq N$ implies $\|\mathbf{x}_{f(k)} - \mathbf{x}\| < \varepsilon$. This means that $B_\varepsilon(\mathbf{x})$ contains $\mathbf{x}_{f(k)}$ for all $k \geq N$, so $B_\varepsilon(\mathbf{x})$ contains $\mathbf{x}_k$ for infinitely many k.

Conversely, suppose that for any $\varepsilon > 0$, the open ball $B_\varepsilon(\mathbf{x})$ contains $\mathbf{x}_k$ for infinitely many k. Thus, for $\varepsilon = 1$, there exists $f(1) \in \mathbb{N}$ with $\mathbf{x}_{f(1)} \in B_1(\mathbf{x})$. Suppose that for all $j = 1, \ldots, m$, we have chosen $\mathbf{x}_{f(j)}$ such that $\mathbf{x}_{f(j)} \in B_{\frac{1}{j}}(\mathbf{x})$, and such that $1 \leq f(1) < f(2) < \cdots < f(m)$. Then with $\varepsilon = \frac{1}{m+1}$, we know $\mathbf{x}_k \in B_{\frac{1}{m+1}}(\mathbf{x})$ for infinitely many k, so we can choose $k = f(m+1) > f(m)$ such that $\mathbf{x}_{f(m+1)} \in B_{\frac{1}{m+1}}(\mathbf{x})$. Thus, by induction, we have constructed a subsequence $(\mathbf{x}_{f(k)})$ of $(\mathbf{x}_k)$, with $\|\mathbf{x}_{f(k)} - \mathbf{x}\| < \frac{1}{k}$. Letting $k \to \infty$, we conclude that $\lim_{k\to\infty} \mathbf{x}_{f(k)} = \mathbf{x}$.

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[7] We know that a nonconvergent sequence can have subsequences which can converge to *different* limits. Give an example of a sequence in $\mathbb{R}^n$ which has *no convergent subsequences* at all. Justify your answer. *Hint:* Recall the Bolzano–Weierstrass Theorem.

Solution. There are many possible answers. Note that the Bolzano–Weierstrass theorem says that any bounded sequence always has a convergent subsequence. Hence we need to consider unbounded sequences. Consider the sequence $(\mathbf{x}_k)$ in $\mathbb{R}^n$ where $\mathbf{x}_k = (k, k, \ldots, k)$. Then $\|\mathbf{x}_k\| = \sqrt{n}k \to \infty$ as $k \to \infty$, so $(\mathbf{x}_k)$ is unbounded. Now let $(\mathbf{x}_{f(k)})$ be any subsequence. Since $f(k) \geq k$, we have $\|\mathbf{x}_{f(k)}\| \geq \sqrt{n}k$ for all k. This is unbounded. So $(\mathbf{x}_{f(k)})$ cannot converge, because any convergent sequence is necessarily bounded. Since $(\mathbf{x}_{f(k)})$ is an arbitrary subsequence, the original sequence $(\mathbf{x}_k)$ has no convergent subsequences.

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[8] Recall that a subset K of $\mathbb{R}^n$ is called *compact* iff every sequence in K admits a subsequence that converges to a point in K.

Let $K_1, K_2, \ldots, K_m$ be a finite collection of compact subsets of $\mathbb{R}^n$. Give two different proofs that the union $\cup_{j=1}^m K_j$ is compact:

[a] using the definition of compactness.

[b] using the Heine–Borel theorem.

Solution.

[a] Let $K = \cup_{j=1}^m K_j$. Suppose that $(\mathbf{x}_k)$ is a sequence in K. We need to show that is admits a subsequence that converges in K. Since we only have finitely many K_j, there must exist at least one $j_0 \in \{1, \ldots, m\}$ such that K_{j_0} contains $\mathbf{x}_k$ for infinitely many values of k. Let $f\colon \mathbb{N} \to \mathbb{N}$ be a strictly increasing function such that $\mathbf{x}_{f(k)} \in K_{j_0}$ for all $k \in \mathbb{N}$. (This exists because K_{j_0} contains $\mathbf{x}_k$ for infinitely many values of k.) Thus $(\mathbf{x}_{f(k)})$ is a subsequence of $(\mathbf{x}_k)$, all of whose terms lie in K_{j_0}, which is compact. Hence there exists a further subsequence $(\mathbf{x}_{f(h(k))})$ that converges to a limit in $K_{j_0} \subseteq K$, so K is compact.

[b] Since each K_j is compact, each K_j is closed and bounded by the Heine–Borel Theorem. Then $K = \cup_{j=1}^m K_j$ is closed since a finite union of closed sets is closed. For every $1 \leq j \leq m$, K_j is bounded implies that there exists some $M_j > 0$ such that $\|x\| \leq M_j$ for all $x \in K_j$. Let $M = \max_{1 \leq j \leq m} M_j$. Then for every $x \in K$, there exists some $1 \leq j \leq m$ such that $x \in K_j$ so $\|x\| \leq M_j \leq M$. Thus K is bounded. Therefore K is compact by the Heine–Borel Theorem.

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[9] Let $E \subseteq \mathbb{R}^n$ be nonempty. For $\mathbf{x} \in \mathbb{R}^n$, we define the *distance* from E to $\mathbf{x}$, denoted $\operatorname{dist}_E(\mathbf{x})$, to be

$$\operatorname{dist}_E(\mathbf{x}) = \inf\{\|\mathbf{x} - \mathbf{z}\| : \mathbf{z} \in E\}.$$

Thus we can think of dist_E as a function from $\mathbb{R}^n$ to $\mathbb{R}$.

[a] Prove that $|\operatorname{dist}_E(\mathbf{x}) - \operatorname{dist}_E(\mathbf{y})| \leq \|\mathbf{x} - \mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.

[b] Prove that $\operatorname{dist}_E \colon \mathbb{R}^n \to \mathbb{R}$ is continuous on $\mathbb{R}^n$.

Solution.

[a] Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and let $\mathbf{z} \in E$. By the triangle inequality, we have

$$\|\mathbf{x} - \mathbf{z}\| \leq \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y} - \mathbf{z}\|.$$

By definition of the distance, $\operatorname{dist}_E(\mathbf{x}) \leq \|\mathbf{x} - \mathbf{z}\|$, so

$$\operatorname{dist}_E(\mathbf{x}) \leq \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y} - \mathbf{z}\|.$$

Thus $\operatorname{dist}_E(\mathbf{x}) - \|\mathbf{x} - \mathbf{y}\|$ is a lower bound of $\{\|\mathbf{y} - \mathbf{z}\| : \mathbf{z} \in E\}$. Since $\operatorname{dist}_E(\mathbf{y})$ is the greatest lower bound, we get $\operatorname{dist}_E(\mathbf{x}) - \|\mathbf{x} - \mathbf{y}\| \leq \operatorname{dist}_E(\mathbf{y})$, and hence

$$\operatorname{dist}_E(\mathbf{x}) - \operatorname{dist}_E(\mathbf{y}) \leq \|\mathbf{x} - \mathbf{y}\|.$$

Interchanging the roles of $\mathbf{x}$ and $\mathbf{y}$, we also have

$$-(\operatorname{dist}_E(\mathbf{x}) - \operatorname{dist}_E(\mathbf{y})) = \operatorname{dist}_E(\mathbf{y}) - \operatorname{dist}_E(\mathbf{x}) \leq \|\mathbf{y} - \mathbf{x}\| = \|\mathbf{x} - \mathbf{y}\|.$$

Putting these together gives

$$|\operatorname{dist}_E(\mathbf{x}) - \operatorname{dist}_E(\mathbf{y})| \leq \|\mathbf{x} - \mathbf{y}\|,$$

as required.

[b] Fix $\mathbf{x}_0 \in \mathbb{R}^n$. Let $\varepsilon > 0$. Choose $\delta = \varepsilon$. If $\|\mathbf{x} - \mathbf{x}_0\| < \delta$, then by part [a] we have

$$|\operatorname{dist}_E(\mathbf{x}) - \operatorname{dist}_E(\mathbf{x}_0)| \leq \|\mathbf{x} - \mathbf{x}_0\| < \delta = \varepsilon.$$

Thus dist_E is continuous at $\mathbf{x}_0$. Since $\mathbf{x}_0$ is arbitrary, we conclude that dist_E is continuous on $\mathbb{R}^n$.

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[10] Let $f \colon \mathbb{R} \to \mathbb{R}$ be given by

$$f(x) = \begin{cases} 0 & \text{if } x \in \mathbb{Q}, \\ 1 & \text{if } x \notin \mathbb{Q}. \end{cases}.$$

Prove that f is not continuous at x for any $x \in \mathbb{R}$.

Hint: Find a way to use the sequential characterization of continuity.

Solution. We proved in class and on a previous assignment, that between any two distinct real numbers there exists a rational number, and there also exists an irrational number. That immediately implies that for any $x \in \mathbb{R}$, there exists a sequence (x_k) of *rational* numbers with $x_k \to x$ as $k \to \infty$ (take x_k rational in $(x - \frac{1}{k}, x + \frac{1}{k})$ to see this) and there also exists a sequence (x_k) of *irrational* numbers with $x_k \to x$ as $k \to \infty$ (take x_k itrational in $(x - \frac{1}{k}, x + \frac{1}{k})$ to see this).

Now let $x \in \mathbb{R}$. If $x \in \mathbb{Q}$, let (x_k) be a sequence of *irrationals* converging to x. Then $f(x) = 0$ but $f(x_k) = 1$ for all k, so $f(x_k)$ does not converge to $f(x)$. Hence f is not continuous at x. If $x \notin \mathbb{Q}$, let (x_k) be a sequence of *rationals* converging to x. Then $f(x) = 1$ but $f(x_k) = 0$ for all k, so $f(x_k)$ does not converge to $f(x)$. Hence f is not continuous at x.
